

Abstract

This study examines the influence of clinical and lifestyle factors from the Framingham Heart Study on the development of coronary heart disease (CHD) over a ten-year period. Using a comprehensive, publicly available dataset, we conducted exploratory data analysis (EDA) to manage missing and outlier data, explore variable relationships, and apply both parametric and non-parametric methods. The study provides a foundation for building robust risk models and identifies significant predictors of CHD, which are crucial for early detection and prevention efforts. The focus is on EDA and not on the modeling itself.

Disclaimer

The paper was not a required part of the term project submission. Therefore, in order to incorporate it into the portfolio, I used OpenAI's ChatGPT 5 model to review and co-author this paper using the codebook. I have reviewed the content to ensure that it accurately outlines the EDA and research performed for the project.

Modeling Cardiovascular Risk: Exploratory Data Analysis Using the Framingham Heart Study

Business Problem Statement

Cardiovascular disease remains the leading cause of morbidity and mortality worldwide. A central challenge in public health is determining which clinical and behavioral factors are most predictive of future CHD, enabling more effective preventive interventions and policy measures. The Framingham dataset provides a unique opportunity to empirically analyze these relationships and assess how modifiable and non-modifiable factors contribute to cardiovascular outcomes.

The analysis centers on several guiding questions: which continuous and categorical variables most influence ten-year CHD risk, how data cleaning and imputation choices affect predictive insights, and whether it is possible to quantify multicollinearity and the independent contribution of each predictor.

Background

The Framingham Heart Study (FHS) is a landmark longitudinal cohort study launched in 1948 to identify common risk factors for cardiovascular disease. The dataset used in this analysis provides a research-friendly snapshot of that broader study, encompassing a range of demographic, clinical, and lifestyle variables. These include age, blood pressure, cholesterol, body mass index (BMI), glucose, heart rate, smoking status, diabetes, hypertension, and medication use. This dataset enables a broad investigation into the multifactorial nature of cardiovascular risk and supports educational exploration of statistical and epidemiological principles.

Data Explanation

Dataset Source and Structure

The dataset contains 4,240 observations and 16 variables after cleaning. These variables span three general categories. Categorical and binary variables include gender, smoking status, blood pressure medication use, prior stroke, hypertension, diabetes, and the ten-year CHD outcome (the target variable). Discrete numerical variables, such as the number of cigarettes per day and heart rate, offer quantifiable measures of behavioral and physiological aspects. Continuous variables—including age, total cholesterol, systolic and diastolic blood pressure, BMI, and glucose—capture the clinical indicators most

associated with cardiovascular risk. See Appendix A for detailed descriptions of each column in the dataset.

Handling Missing and Outlier Data

Missing values were most common in glucose and total cholesterol. For normally distributed variables, the mean was used for imputation; for skewed distributions, the median was applied. Outlier detection and management were addressed through both log transformation and interquartile range (IQR) filtering. However, removing outliers did not yield meaningful improvements in predictive performance, so the default approach retained all data for analysis.

Methods

Exploratory Data Analysis

EDA was performed through visual and statistical techniques. Distributions of continuous variables were examined using histograms and boxplots, which revealed the data's shape, spread, and potential outliers. Patterns of missingness were mapped to guide imputation strategies. Pairwise relationships were evaluated using correlation matrices and pair plots, providing a preliminary view of the linear associations among predictors and the target variable.

Outlier Analysis

Correlation matrices were computed before and after outlier removal to assess the stability of relationships. Both log transformation and IQR filtering were applied to determine whether outlier treatment meaningfully altered correlation strength or regression results.

PMF and CDF Analysis

Probability Mass Functions (PMFs) were used to compare the distributions of key variables between individuals who developed CHD and those who did not. This analysis revealed that higher age and BMI were strongly associated with CHD. Cumulative Distribution Functions (CDFs) further illustrated these group differences, emphasizing how risk accumulates across variable ranges.

Hypothesis Testing and Regression

Group differences were confirmed using mean and permutation tests. At the same time, both simple and multiple linear regression models quantified the independent contributions of variables, such as blood pressure and glucose, to the risk of CHD. These analyses validated many of the expected risk factor relationships established in cardiovascular literature.

Multicollinearity Assessment

Variance Inflation Factors (VIFs) were calculated for each predictor in the logistic regression framework to ensure that intercorrelated variables did not distort results. All VIF values were below five, indicating no evidence of problematic multicollinearity. Variables significant at the 0.05 level were retained for subsequent modeling, while non-significant predictors were flagged for potential exclusion.

Key Results

The analysis consistently identified age, systolic blood pressure, and glucose as significant predictors of ten-year CHD risk. PMF and CDF analyses showed that older individuals and those with higher BMI faced notably greater risk. Additional factors such as prior stroke, hypertension, diabetes, and smoking contributed minor but still meaningful effects. The absence of multicollinearity confirmed the robustness of model estimates, though R^2 values were modest, reflecting the inherent difficulty of clinical prediction in diverse population-based cohorts.

Limitations

The Kaggle version of the Framingham dataset omits several original study variables, including HDL and LDL cholesterol fractions, as well as family history, which limits the depth of risk stratification. Imputation methods introduce uncertainty, particularly for glucose, where missing data were substantial. Furthermore, this analysis did not explore complex nonlinear relationships or variable interactions. Future studies employing advanced statistical and machine learning methods could address these gaps to improve predictive accuracy.

Future Applications

Expanding the scope of variables to include genetic, behavioral, or lifestyle factors could enhance CHD risk discrimination. Modern machine learning methods such as random forests, gradient boosting, and regularized regression (Lasso and Ridge) offer promising avenues for model refinement. Additionally, feature engineering—such as decomposing BMI into separate height and weight predictors—may yield richer insights into the role of body composition in cardiovascular health.

Ethical Assessment

The dataset used is de-identified and publicly accessible, ensuring participant privacy. However, as modeling efforts evolve toward integration with clinical data, it will be critical to maintain strong ethical standards regarding data governance and informed consent.

Notably, model predictions should support, not replace, professional clinical judgment in assessing and managing individual cardiovascular risk.

Conclusion

This exploratory data analysis of the Framingham Heart Study reaffirms well-established cardiovascular risk factors and demonstrates how a structured, reproducible EDA process can inform subsequent predictive modeling. While preprocessing decisions—such as imputation and outlier handling—affect minor aspects of the analysis, core predictors remain consistent across methodologies. The study highlights the enduring value of the Framingham data for teaching, research, and the continuing development of predictive frameworks in epidemiological health science.

Appendix A: Variables and Definitions

Variable	Description	Type
male	Gender (1=male, 0=female)	Categorical
age	Age (years)	Continuous
education	Education level (details not specified)	Numeric
currentSmoker	Currently smokes (1=yes, 0=no)	Categorical
cigsPerDay	Cigarettes smoked per day	Numeric
BPMeds	On blood pressure medication (1=yes, 0=no)	Categorical
prevalentStroke	History of stroke (1=yes, 0=no)	Categorical
prevalentHyp	History of hypertension (1=yes, 0=no)	Categorical
diabetes	Diabetes diagnosis (1=yes, 0=no)	Categorical
totChol	Total cholesterol (mg/dL)	Continuous
sysBP	Systolic blood pressure (mmHg)	Continuous
diaBP	Diastolic blood pressure (mmHg)	Continuous
BMI	Body mass index (kg/m ²)	Continuous
heartRate	Pulse at exam	Numeric
glucose	Fasting blood glucose (mg/dL)	Continuous
TenYearCHD	Developed CHD in 10 years (1=yes, 0=no)	Target

References

Downey, A. B. (2014). *Think Stats: Exploratory Data Analysis in Python* (2nd ed.). O'Reilly Media. ISBN: 9781491907337.

Aasheesh. (2022). *Framingham Heart Study dataset* [Data set]. Kaggle. Framingham Heart Study dataset.